

Name

ID#

Math 150 Spring 2025 Final Exam

- Desmos is allowed (and the only app website allowed - no other apps or websites are allowed at anytime)
- Must show all work to receive credit
- All integrals must be
- Attach all scratch paper

	Question	Points
1	Integration	1 / 5
2	Riemann Sums	2 / 5
3	Volume	1 / 5
4	Limits & Continuity	4 / 4
5	Tangent & Normal Lines	1 / 4
6	Surface Area	0 / 4
7	Optimization	0 / 4
8	Related Rates	1 / 3
9	First Derivative Test	/ 8
10	Geometry	/ 3
Total		10 / 40

choose 10 of 3

1) Evaluate the integral $\int 43 \tan^5(x) \sec^4(x) dx$ using two different substitutions and discuss how the two anti-derivatives are related.

$$\int 43 \tan^5 x \sec^4 x dx$$

$$u = \tan x$$

$$du = \sec^2 x$$

$$43 \int \tan^5 x \sec^4 x dx$$

$$u^5$$

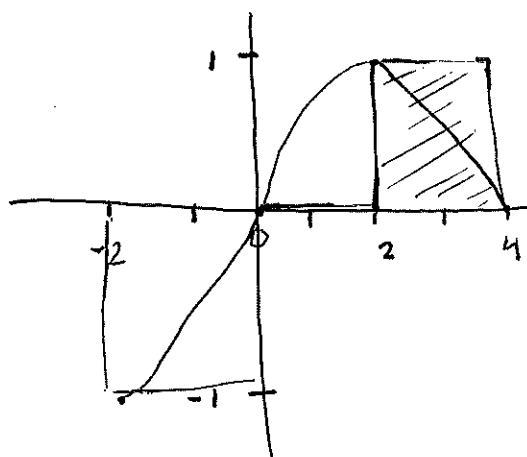
Name _____ ID# _____

2) Estimate the signed area under the curve $y = \sin\left(\frac{\pi x}{4}\right)$ from -2 to 4 using:

- (a) three (3) rectangles with the **left endpoint** approximation.
- (b) three (3) rectangles with the **right endpoint** approximation,
- (c) three (3) rectangles with the **midpoint** approximation,
- (d) Find the exact signed area using FTC and compare your estimates from parts (a-c)

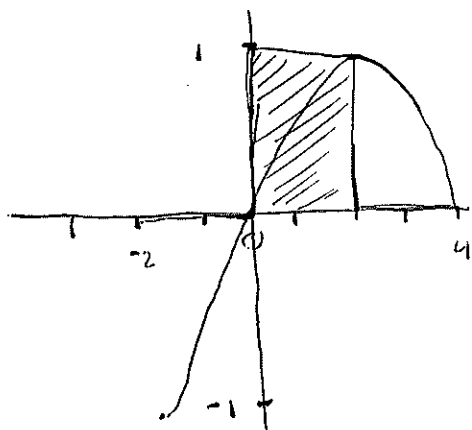
In each case a-c draw the graph and rectangles and use sigma notation.

a)



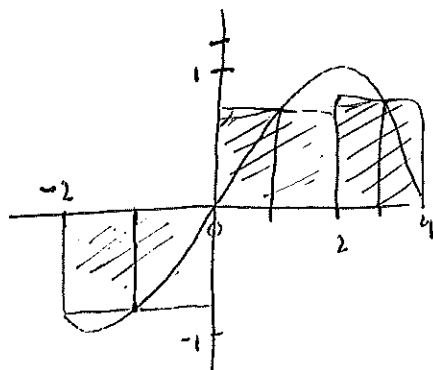
Σ

b)



Σ

c)



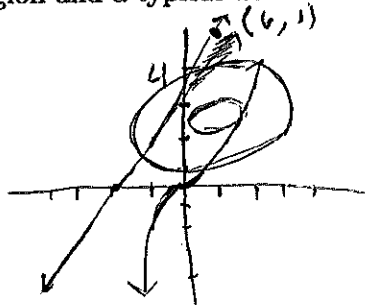
Σ

3) Find the volume generated by revolving the region bounded by the curves

$$y = 2x + 4, x = 1, y = x^3$$

about the y -axis first using washers, then using shells.

Draw the region and a typical washer. Draw the region and a typical shell.



Find the exact value of volume using washers.

Find the exact value of volume using shells.

Name _____ ID# _____

4) Sketch the graph of a function with following limits, and classify any discontinuities.

- $\lim_{x \rightarrow -2^-} f(x) = -2$

- $\lim_{x \rightarrow -2^+} f(x) = 1$

- $f(-2) = 3$

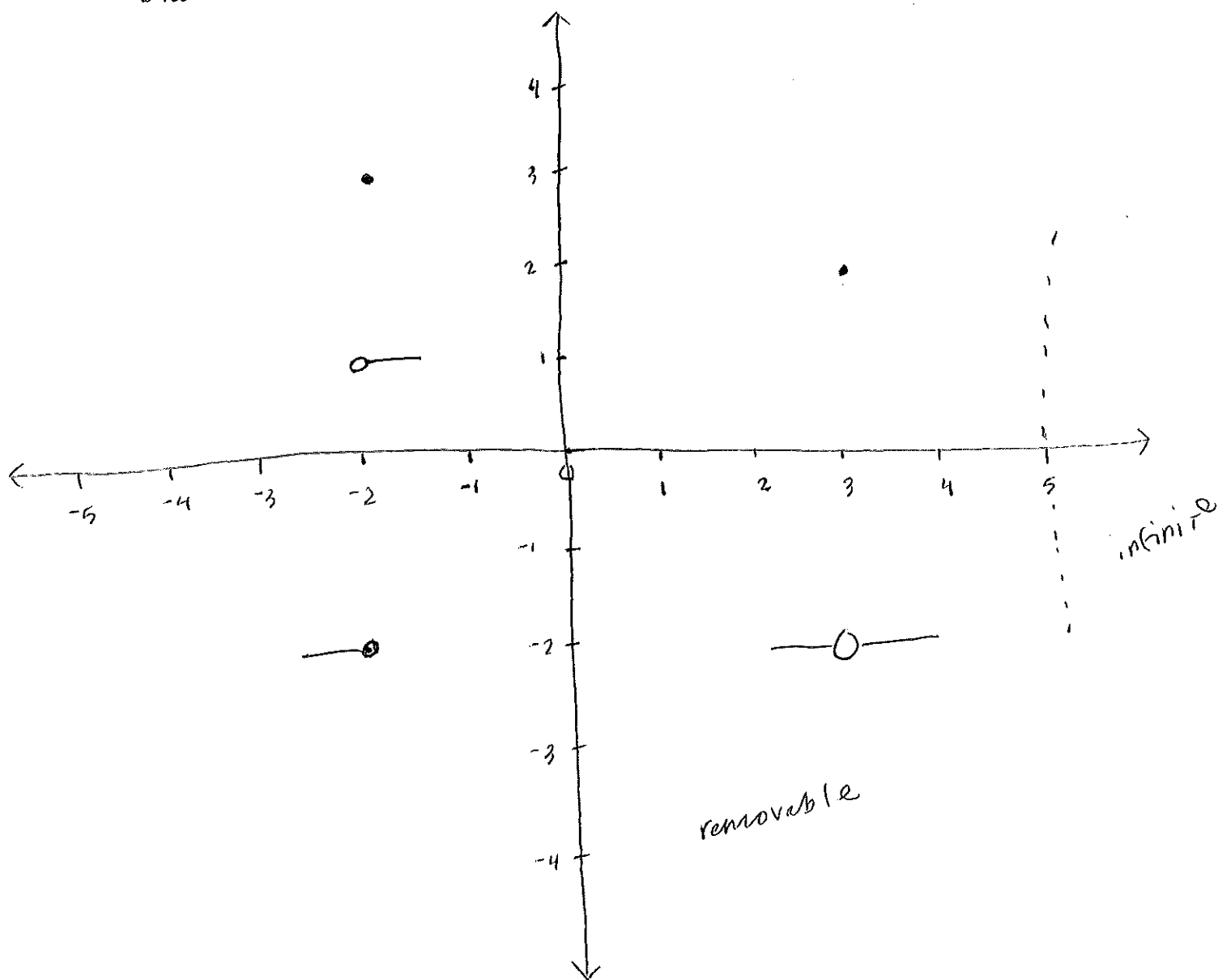
- $\lim_{x \rightarrow 3} f(x) = -2$

- $f(3) = 2$

- $\lim_{x \rightarrow 5^+} f(x) = \infty$

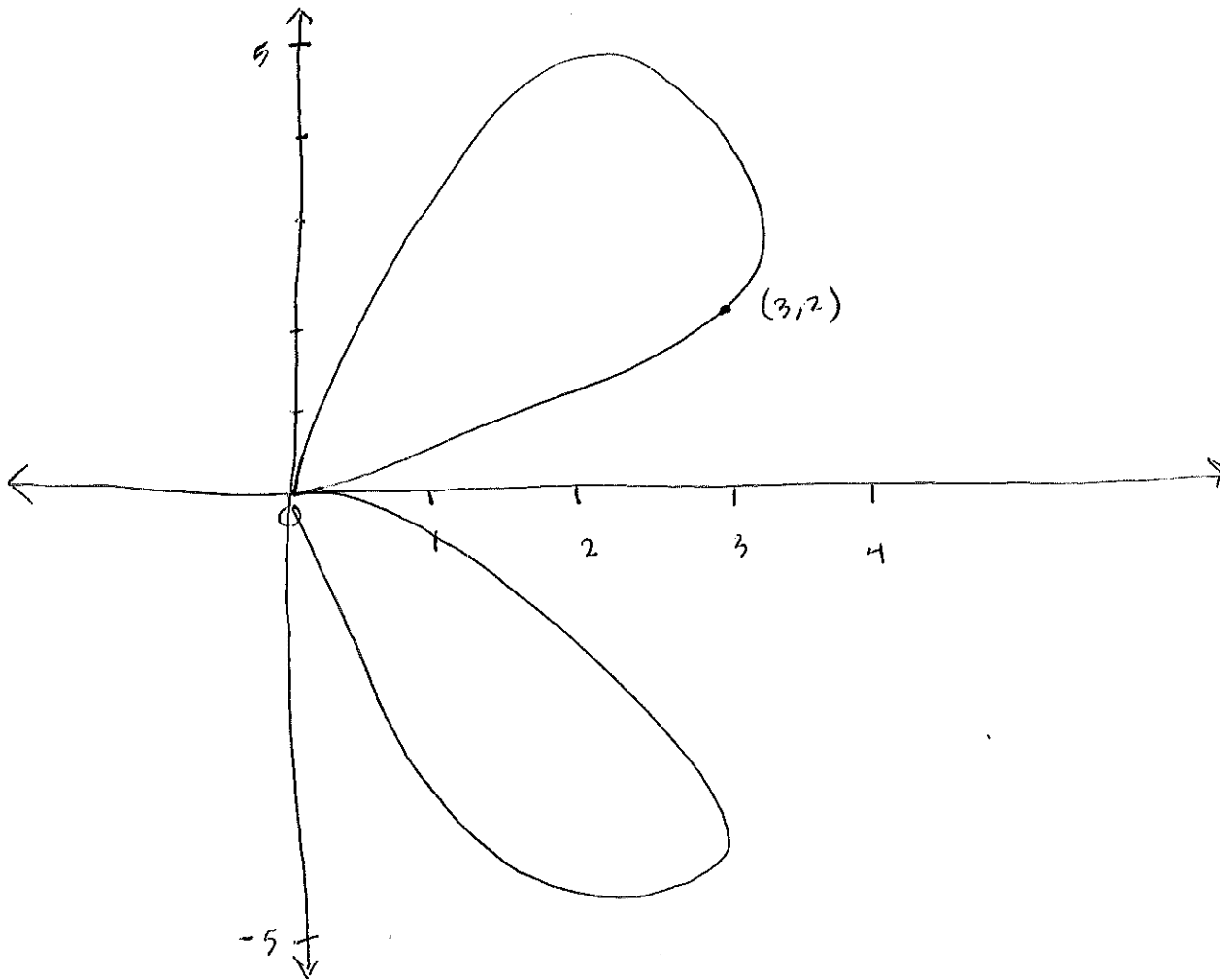
- $\lim_{x \rightarrow 5^-} f(x) = \infty$

- $\lim_{x \rightarrow \infty} f(x) = 1$



5) Graph the equation $12(x^2 + y^2)^2 = 169xy^2$

Find the tangent line and normal line at the point $(3, 2)$



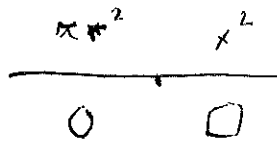
Name _____ ID# _____

6) Calculate the surface area of the surface obtained by revolving the curve $y = \frac{x^3}{3}$ around the x -axis from $1 \leq x \leq 2$.

I don't remember anything :-

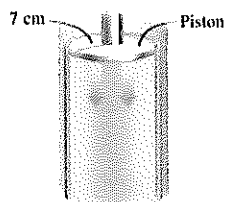
I'm never taking a calculus class
again. Please pass me.

7) A wire of length 100 mm is cut into two pieces, forming a circle with the first segment, and a square with the remaining segment, while optimizing the sum of the area of the circle and square. State the objective function and the interval of interest.



Name _____ ID# _____

8) A piston is seated at the top of a cylindrical chamber with radius 7cm when it starts moving into the chamber at a constant speed of 3 cm / sec. (see figure).



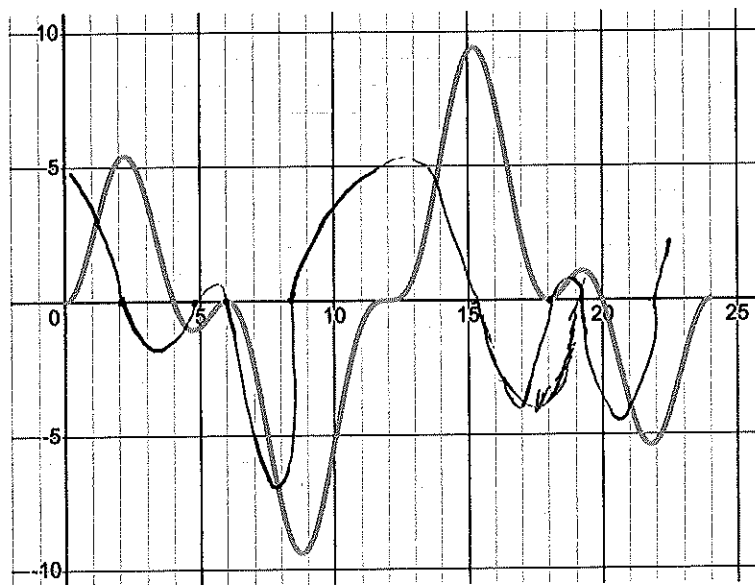
$$\pi 7^2 \cdot h$$

$$V = \pi r^2 \cdot L$$

What is the rate of change of the volume of the cylinder when the piston is 9 cm from the base of the chamber?

9) Given the graph of power $P(t)$ over a 24 hour period below, and that power $P(t) = E'(t)$.

Graph energy $E(t)$ and estimate the time(s) that **energy** is maximized and minimized.
(Hint use the first derivative test)



maximized: ~ 3 pm
minimized: ~ 8 am

Name _____ ID# _____

10) Find the exact value of $\int_{-5}^0 \sqrt{25 - x^2} \, dx$ using geometry.

Find the exact value of $\int_{-5}^0 \sqrt{25 - x^2} \, dx$ using integration techniques.